

MLT-04 How to Develop a Trading Strategy using Machine Learning Summary

Key Takeaways

- Using Python and Jupyter notebooks to create features.
- Using Python and Jupyter notebooks to build and evaluate models, use feature selection and test raw performance

Pre-requisites:

- Understanding Python functions, control flow statements (if-else-continue), loops, list comprehensions.
- Familiarity with basic operations using Pandas dataframes and series.
- Know how to create a virtual conda environment from a yaml file (can be found online).
- Familiarity using Jupyter notebooks.
- Understanding the basics of machine learning including random forest, gradient boosting, bagging, feature selection, model selection, model validation.

Downloadable Files

- MLT-04-Files.zip

Post this lecture, you will be in a position to:

- Create a set of features for different timeframes
- Use Python to investigate and analyze the features
- Use Python to build and evaluate ML models for a potential trading strategy
- Understand the basics of information bars, and how to create them
- Incorporate information bars to build better models
- Raw summary backtest the strategy using Python and Pyfolio

These concepts are utilized to help you utilize machine learning specifically (tree-based models) to develop a trading strategy. The primary tools used are Python, and Jupyter Notebooks.

Additional Readings

<https://news.cornell.edu/stories/2020/08/study-machine-learning-can-predict-market-behavior>
<https://towardsdatascience.com/advanced-candlesticks-for-machine-learning-i-tick-bars-a8b93728b4c5>
<https://towardsdatascience.com/advanced-candlesticks-for-machine-learning-ii-volume-and-dollar-bars-6cda27e3201d>
<https://towardsdatascience.com/financial-machine-learning-part-0-bars-745897d4e4ba>
<https://christophm.github.io/interpretable-ml-book/shapley.html>
https://shap.readthedocs.io/en/latest/example_notebooks/overviews/An%20introduction%20to%20explainable%20AI%20with%20Shapley%20values.html
<https://arxiv.org/pdf/2202.05594>
<https://www.analyticsvidhya.com/blog/2019/11/shapley-value-machine-learning-interpretability-game-theory/>
<https://neptune.ai/>
<https://arxiv.org/abs/2005.12483> <- E.P. Chan paper
https://scikit-learn.org/stable/modules/generated/sklearn.metrics.matthews_corrcoef.html

https://scikit-learn.org/stable/modules/generated/sklearn.metrics.precision_recall_curve.html
<https://arxiv.org/abs/2005.12483>
<https://docs.conda.io/projects/conda/en/stable/user-guide/getting-started.html>
<https://hudsonthames.org/fractional-differentiation>
<https://medium.com/swlh/fractionally-differentiated-features-9c1947ed2b55>
<https://docs.conda.io/projects/conda/en/stable/user-guide/tasks/manage-environments.html>
[#creating-an-environment-from-an-environment-yml-file](#)

Recommended time for the session & related coursework - 7 hours